

Reweighting MNLI Using Hypothesis-Only Difficulty Estimates: A Study of Importance Weighting for NLI Robustness

Abstract

Natural Language Inference (NLI) datasets contain annotation artifacts that enable *hypothesis-only* classifiers to perform well above majority-class baselines. A natural question is whether full NLI models can be improved by downweighting training examples that appear “easy” for such heuristic-based models. We train a hypothesis-only classifier on SNLI, apply it to MNLI to estimate example difficulty, and construct three reweighted training sets that downweight easy examples. We evaluate ELECTRA-based NLI models trained on these weighted datasets. Across three reweighting schemes and three random seeds, we find no improvements over a standard MNLI baseline; all weighted models perform slightly worse, with differences within seed variance. Our analysis suggests that hypothesis-only confidence is a weak and noisy difficulty signal, and that simple importance weighting does not meaningfully improve NLI robustness.

1 Introduction

Large NLI datasets such as SNLI (Bowman et al., 2015) and MNLI (Williams et al., 2018) have been shown to contain annotation artifacts that allow “hypothesis-only” models to achieve surprisingly high accuracy (Poliak et al., 2018). These models exploit lexical cues rather than performing true inference, indicating that much of the training data may be trivially solvable.

This motivates a simple idea: if some training examples are “easy” for a hypothesis-only model, perhaps they contribute little to learning robust representations. Reweighting the dataset to emphasize “harder” examples may encourage a full NLI model to rely less on artifacts and more on deeper semantic reasoning.

To test this hypothesis, we train a hypothesis-only teacher model on SNLI and apply it to

MNLI to estimate per-example difficulty using the teacher’s confidence. We construct three reweighted training sets—a random-weight control, a confidence-based weighting, and a rescaled version preserving mean weight—and train ELECTRA-small models on each variant.

We evaluate these models on MNLI and analyze their disagreements and error patterns. Despite multiple reweighting strategies and careful controls, we observe no improvements over the standard baseline. In fact, each weighted model achieves slightly worse accuracy. We conclude that hypothesis-only confidence is not a reliable difficulty signal for importance weighting in NLI.

2 Approach

2.1 Data Sources

We use MNLI as the target dataset for model training and evaluation, and SNLI for training a hypothesis-only teacher model. Experiments use the MNLI matched validation set.

2.2 Teacher Model

It would not be valid to use an MNLI hypothesis-only trained model to rate the MNLI training set, because it could simply memorize the answers. Instead, we train an ELECTRA-small-discriminator model on SNLI, providing only the hypothesis. It achieves 68.1% accuracy on the SNLI development set which closely matches the accuracy achieved in previous work (Poliak et al., 2018). This model achieves 46.3% accuracy when applied to MNLI, which suggests that the artifacts in one set do not transfer fully to another. Still, this model provides a noisy but workable difficulty estimate.

Table 1 shows how teacher-assigned difficulty categories map to MNLI genres. The distributions

are relatively uniform, indicating that teacher difficulty correlates only weakly with genre.

Category	Fiction	Govt	Travel	Slate	Tel.
Very easy	23.6	17.5	20.6	16.6	21.7
Easy	18.8	22.0	19.0	19.4	20.8
Medium	18.0	20.8	22.0	18.5	20.7
Hard	19.8	18.7	19.5	20.6	21.5

Table 1: Genre distribution across teacher difficulty categories.

2.3 Reweighting Methods

We explore three schemes for assigning per-example weights w_i . Let c_i denote the teacher logit confidence for example i . Figure 1 shows the distributions for each scheme

Reweighted (v1). Incorrect examples receive full weight, while correct examples are down-weighted proportionally to the teacher model’s confidence:

$$w_i = \begin{cases} 1.0, & \text{if incorrect} \\ \frac{c_i - \min(c)}{\max(c) - \min(c) + \epsilon}, & \text{if correct.} \end{cases}$$

Reweighted + Rescaled (v2). We rescale v1 weights so the mean is exactly 1.0:

$$w'_i = \frac{w_i}{\frac{1}{N} \sum_j w_j}.$$

Random (v3). We assign full weight to all examples the teacher model predicts incorrectly and randomly downweight all examples the teacher model predicts correctly:

$$w_i = \begin{cases} 1.0, & \text{if incorrect} \\ \max(\mathcal{N}(0.5, 0.2^2), \epsilon), & \text{if correct.} \end{cases}$$

Weights are applied by multiplying each example’s cross-entropy loss during training. Validation and test losses are unweighted.

3 Experimental Setup

All models use google/electra-small-discriminator. Training uses the HuggingFace Trainer (MPS hardware), with identical hyperparameters for all runs. Each model is trained with three random seeds.

We evaluate:

- Baseline (unweighted MNLI),
- Reweighted v1,
- Rescaled v2,
- Random-weight v3 (control).

4 Results

4.1 Overall MNLI Performance

Table 2 shows accuracy across seeds. All weighted models perform slightly below the baseline, but within expected variance.

Seed	Base	v1	v2	v3
1	0.8034	0.8019	0.8012	0.8016
2	0.8122	0.7962	0.8001	0.8030
3	0.8049	0.7992	0.8020	0.8054
Mean	0.807	0.799	0.801	0.803
Std	0.005	0.003	0.001	0.002

Table 2: Accuracy on MNLI development set.

4.2 Performance by Teacher-Defined Hardness

Because the goal of importance weighting is to focus the model on more challenging examples, we evaluate model performance separately on MNLI dev examples that the SNLI hypothesis-only teacher model classified as “hard” or “easy.” Hard examples are those the teacher model predicts incorrectly; easy examples are those it predicts correctly with high confidence.

Table 3 shows accuracies for both slices. All weighted models outperform the baseline on the hard subset by approximately 2–3 absolute percentage points. This indicates that the weighting procedures did influence the optimization dynamics in the intended direction: the model places greater emphasis on examples that the teacher model finds difficult.

However, this improvement comes at the cost of substantially reduced accuracy on the easy subset, where all weighted models perform worse than the baseline by 3–4 points. Because easy examples constitute a substantial portion of the MNLI dev set, this degradation dominates the overall accuracy, resulting in slightly lower aggregate performance.

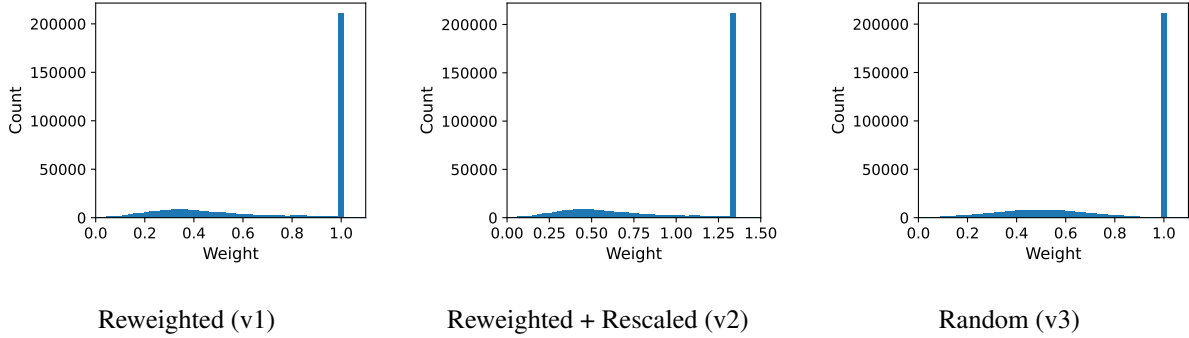


Figure 1: Histograms of weight distributions for all reweighting schemes.

Model	Hard Accuracy	Easy Accuracy
Baseline	0.7602	0.8595
Weighted v1	0.7843	0.8174
Weighted v2	0.7830	0.8235
Weighted v3	0.7837	0.8265

Table 3: Performance on teacher-defined hard vs. easy subsets of MNLI dev. Weighted models improve on hard examples but degrade on easy ones.

4.3 Weighted Model Agreement with Baseline

Across the MNLI development set, all weighted models behave similarly to the baseline model. As shown in Table 4, the predictions of each weighted model agree with the baseline on roughly 90% of examples, indicating that reweighting does not meaningfully alter the model’s overall decision patterns.

On the remaining $\sim 10\%$ of cases where the models disagree, the weighted models are correct only about 42% of the time—roughly equal to chance relative to the baseline. This balanced error profile shows that disagreements are not systematic improvements nor consistent failures, but rather small, noise-level deviations.

Taken together, these results demonstrate that despite applying substantially different training weight distributions, the resulting models converge to almost the same decision function as the unweighted baseline.

Model	Agr.	BothC	BothI	OtherC	BaseC
v1	89.8	75.7	15.3	41.4	4.8
v2	89.5	75.7	15.0	42.7	4.8
v3	89.4	75.7	15.0	43.0	4.7

Table 4: Agreement of weighted models vs. baseline.

4.4 Error Analysis

A core goal of this work is to determine whether importance weighting changes the types of mistakes the MNLI model makes. If weighting successfully shifts the model toward handling more challenging cases, we would expect to see systematic differences in error patterns across lexical cues, genres, or other subpopulations.

To examine this, we perform two complementary analyses: (1) keyword-level error statistics and (2) genre-level error rates.

Identifying high-risk keywords. To analyze whether weighting changes the types of mistakes the model makes, we begin by computing a vocabulary-level profile of which words are most strongly associated with errors. We break the hypothesis field into a bag of words, count how frequently each word appears in correct versus incorrect predictions, and compute two statistics:

$$\text{error_rate}(w) = P(\text{incorrect} \mid w)$$

$$\text{lift}(w) = \frac{\text{error_rate}(w)}{\text{global error rate}}$$

Words with high lift occur disproportionately often in incorrect predictions, making them reliable indicators of linguistic phenomena that challenge the model. From the top portion of this ranked list, we select a representative subset that correspond to well-studied sources of difficulty in NLI.

Keyword-level results. Table 5 reports error rates for selected keywords across all models. They exhibit nearly the same error rates as the baseline. Differences are small (typically < 2 points) and inconsistent across weighting

schemes. Notably, negation terms such as *no*—often a driver of NLI difficulty—show no systematic improvement under any weighting strategy.

Overall, keyword-level analysis indicates that none of the reweighting schemes change how the model processes linguistically meaningful phenomena. The model’s failure patterns are highly stable.

Genre-level analysis. We also analyze performance by MNLI genre. Table 6 shows that genre-level differences across baseline and weighted models are minimal. All models perform best on Government and worst on Slate, reproducing standard MNLI patterns. Weighting affects overall error rate slightly but does not shift genre sensitivity.

Interpretation. These analyses reveal two important facts: (1) The reweighting procedure does *not* cause the model to specialize or improve on specific linguistic phenomena, even those strongly associated with teacher-defined “hard” examples. (2) The model’s error distribution remains almost unchanged across all weighting schemes, despite modifying around 50% of the training weights.

Combined with the earlier result that weighted models perform better on teacher-defined hard examples but worse on easy examples, the error analysis suggests that the weighting signal shifts the global decision boundary slightly but does not meaningfully alter the model’s underlying inductive biases or error signature.

This stability reinforces our conclusion: hypothesis-only confidence is too noisy and too weak a difficulty signal to produce targeted improvements or meaningful redistribution of error behavior.

Model	less	people	don’t	only	but	no
Baseline	27.7	24.8	23.9	23.6	22.9	18.9
v1	29.9	23.1	23.9	22.8	22.9	19.4
v2	29.9	24.1	22.4	24.1	24.1	19.3
v3	32.7	24.7	22.0	20.7	20.7	19.2

Table 5: Error rates on MNLI for samples containing selected keywords.

5 Discussion

Our results demonstrate that hypothesis-only confidence is not a reliable difficulty estimate for importance weighting. We identify several reasons:

Model	Slate	Fiction	Tel.	Travel	Govt.
Baseline	24.9	20.1	19.0	18.9	14.7
v1	25.9	20.9	19.9	18.5	15.3
v2	25.1	20.8	19.4	19.0	15.1
v3	25.0	21.1	19.1	18.0	15.4

Table 6: Error rates by genre on MNLI.

Noise Amplification. Examples misclassified by the teacher often reflect label noise or ambiguity. Upweighting them may harm performance.

Weak Linguistic Signal. Teacher-defined difficulty does not correlate strongly with known “hard” categories such as negation or quantifiers; keyword and genre-based analyses show similar patterns across all models.

Small Weighting Effect. The weight ranges (0–1 or rescaled to mean 1) may be too mild to shift the optimization trajectory. Random-weight controls perform similarly to confidence-based weighting, suggesting that the model is relatively insensitive to these small perturbations.

Dataset Asymmetry. The teacher is trained on SNLI and evaluated on MNLI, introducing cross-dataset mismatch. The resulting difficulty signal may reflect SNLI-specific artifacts that do not transfer cleanly to MNLI.

6 Conclusion

We hypothesized that emphasizing examples that are difficult for a hypothesis-only model would improve NLI performance. Across four training regimes and three random seeds, we observed no improvements over an unweighted baseline. Weighted models performed slightly worse, and although they improved on a teacher-defined hard subset, they degraded on easy examples enough to hurt overall performance. Our findings suggest that naive confidence-based importance weighting is insufficient for debiasing NLI models, and that future work should explore more structured difficulty signals or dynamic curricula.

References

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